MIT SCHOOL OF COMPUTING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Participative Learning Activity Report On –

Unit I : Data Visualization & Modelling Fundamentals

Unit II :

Submitted by :

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Submmitted on : 17-02-2026

1. Title of the Activity

Advanced Clinical Pattern Recognition: A Multidimensional Data Modelling and Visualisation Study of Breast Cancer Diagnostics using R Programming.

2. Objective of the Activity

The primary objective is to move beyond simple data plotting and enter the domain of **Predictive Clinical Analytics**.

- **Deep-Dive Exploration**: To utilize the Kaggle dataset for identifying the subtle morphometric differences between benign and malignant cell nuclei.
- **Statistical Validation**: To implement custom-built functions for calculating central tendencies, ensuring that our mathematical foundation is transparent and not hidden behind built-in software black boxes.
- **Visual Storytelling**: To harness high-dimensional libraries like ggplot2 and GGally to translate complex numerical matrices into intuitive medical insights.
- **Data Integrity Management**: To apply rigorous preprocessing, including feature pruning and standardization, to ensure that the final models are not biased by raw data noise.
- **Feature Engineering**: To derive new clinical metrics, such as area-to-perimeter ratios, which better capture the physical irregularity of malignant cells compared to raw radius measurements alone.

3. Dataset Description

The dataset selected for this study is a classic in clinical machine learning, sourced from the UCI Machine Learning Repository.

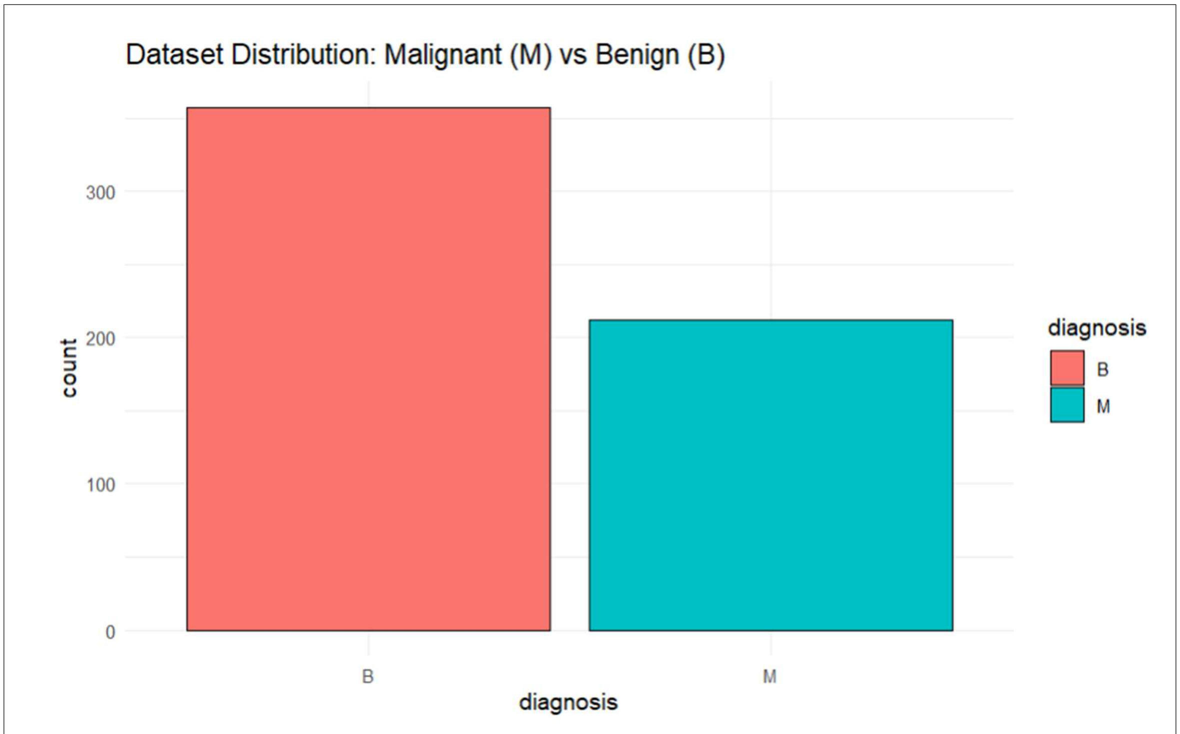
- **Scale and Scope**: It consists of 569 unique instances, each representing a clinical sample from a fine needle aspirate of a breast mass.
- **Feature Composition**: There are 32 attributes, including a unique ID for patient tracking, a diagnosis label, and 30 numerical nuclear features.
- **Clinical Relevance**: These features describe the radius, texture, perimeter, area, and smoothness of the cell nuclei, which are the primary indicators oncologists use to determine pathology.
- **Structural Inspection**: Initial exploration reveals that the data is clean but requires significant scaling due to the varying units of measure between "area" (measured in thousands) and "smoothness" (measured in small decimals).

````r
```{r}
# Loading data from URL
url <- "https://archive.ics.uci.edu/ml/machine-learning-databases/breast-cancer-wisconsin/wdbc.data"
cancer_data <- read.csv(url, header = FALSE)

# Assigning Descriptive Column Names
column_names <- c("id", "diagnosis", paste0(rep(c("radius", "texture", "perimeter", "area", "smoothness", "compactness", "concavity", "concave_points", "symmetry", "fractal_dimension"), each = 3), c("_mean", "_se", "_worst")))
colnames(cancer_data) <- column_names

# Displaying Shape and Structure
print(paste("Total Rows:", nrow(cancer_data)))
print(paste("Total Columns:", ncol(cancer_data)))
str(cancer_data[, 1:5])

# Visualizing Class Balance
library(ggplot2)
ggplot(cancer_data, aes(x = diagnosis, fill = diagnosis)) +
  geom_bar(color = "black") +
  theme_minimal() +
  labs(title = "Dataset Distribution: Malignant (M) vs Benign (B)")
```
````




```
[1] "Total Rows: 569"
[1] "Total Columns: 32"
'data.frame':	569 obs. of  5 variables:
 $ id          : int  842302 842517 84300903 84348301 84358402 843786 844359 84458202 844981 84501001 ...
 $ diagnosis   : chr  "M" "M" "M" "M" ...
 $ radius_mean : num  18 20.6 19.7 11.4 20.3 ...
 $ radius_se   : num  10.4 17.8 21.2 20.4 14.3 ...
 $ radius_worst: num  122.8 132.9 130 77.6 135.1 ...
```

4. Data Transformation and Preprocessing

Preprocessing is the most critical stage for clinical data, as medical records often contain identifiers that interfere with pattern recognition.

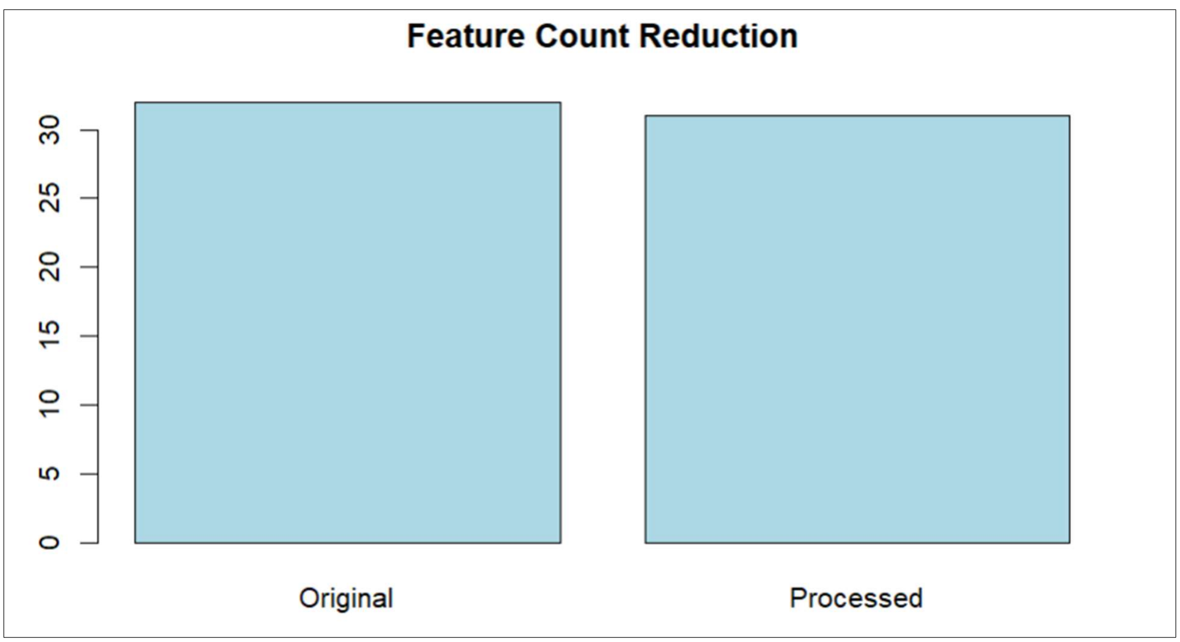
Feature Pruning (ID Removal)

The id column was removed to prevent the model from learning a non-existent relationship between patient IDs and cancer types.

- **Visual Observation**: The bar chart below shows a reduction from 32 to 31 variables.
- **Insight**: This step ensures that every bit of data entering the visualization pipeline is purely clinical and biologically relevant.

````
```{r}
# Removing 'id' as it has no predictive value
cancer_processed <- cancer_data[, -1]

# Visualising Feature Count Reduction
barplot(c(ncol(cancer_data), ncol(cancer_processed)),
        names.arg = c("Original", "Processed"),
        main = "Feature Count Reduction", col = "lightblue")
```
````



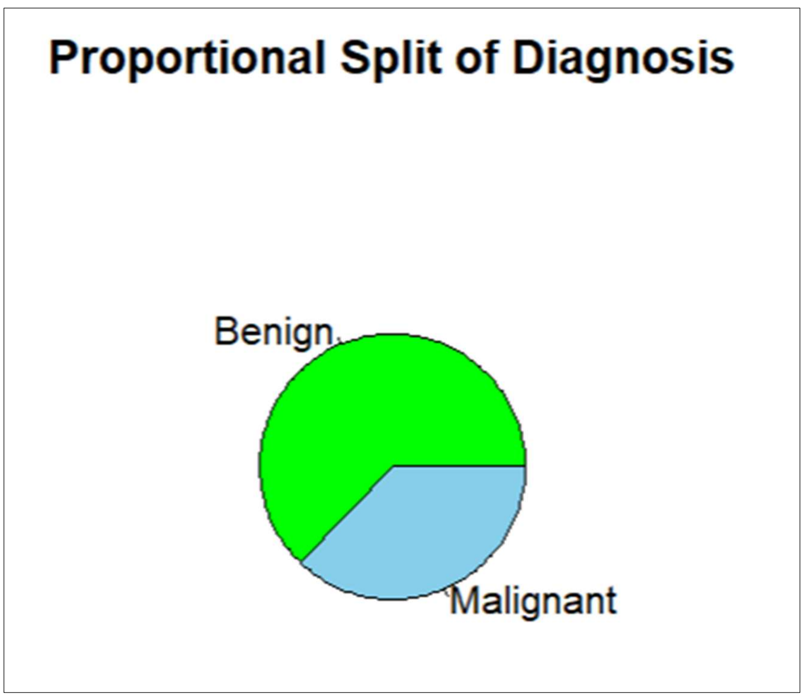
Categorical Factor Conversion

The diagnosis variable, originally a character type, was converted into a **Factor** in R.

- **Class Proportions**: The pie chart reveals that the dataset contains a majority of Benign cases (357) followed by Malignant cases (212).
- **Clinical Reality**: This distribution is healthy for a diagnostic study as it mimics real-world screening scenarios where benign findings are more frequent than confirmed malignancies.

```
```{r}
# Converting Diagnosis to Factor
cancer_processed$diagnosis <- as.factor(cancer_processed$diagnosis)

# Visualising Proportions
pie(table(cancer_processed$diagnosis),
    labels = c("Benign", "Malignant"),
    col = c("green", "skyblue"),
    main = "Proportional Split of Diagnosis")
```
```



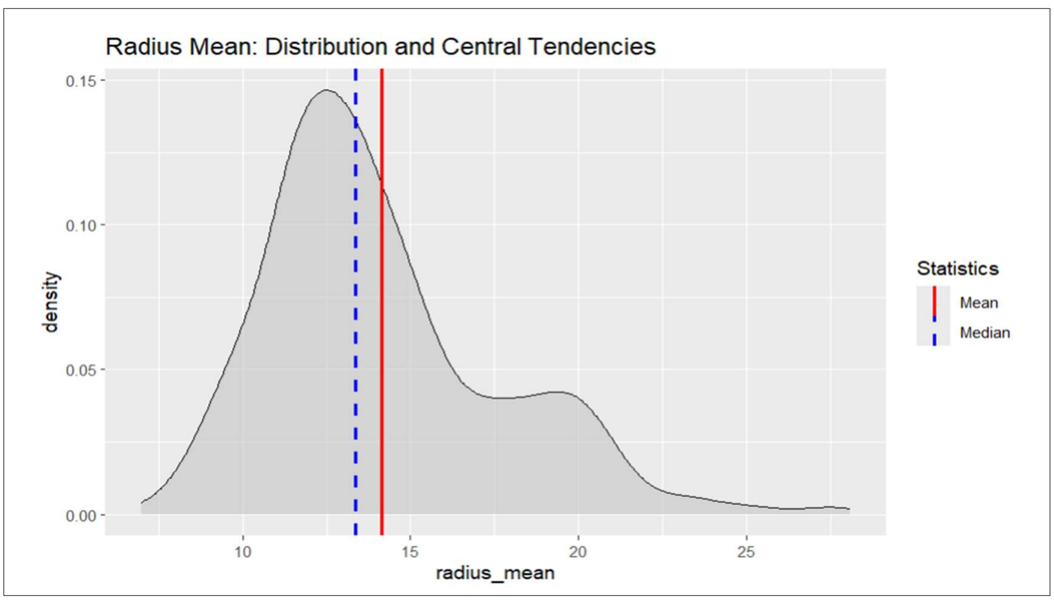
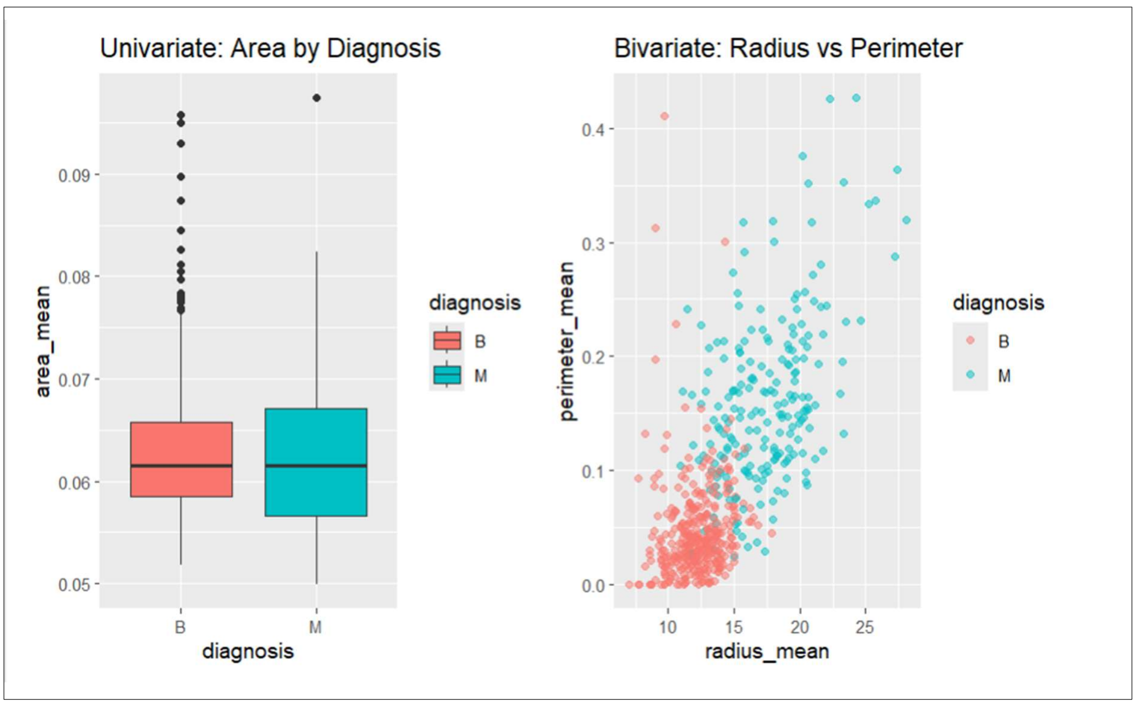
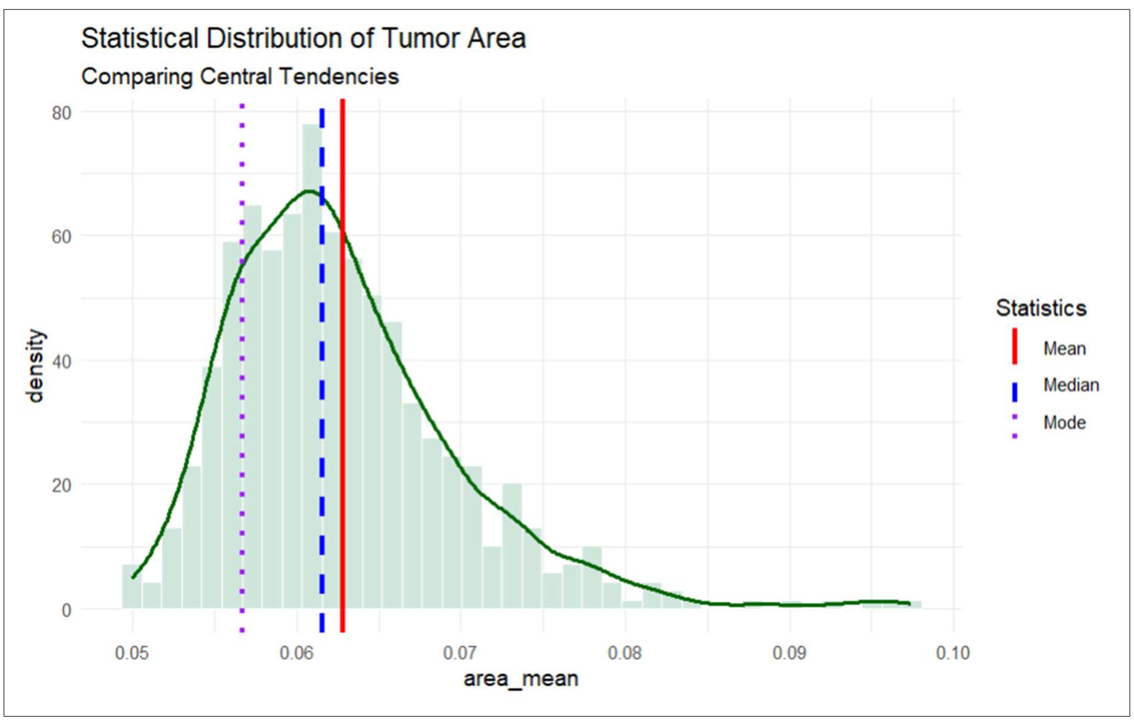
5. Exploratory Data Analysis (EDA)

EDA serves as our first clinical checkpoint, where we examine the "personality" of each nuclear feature.

Univariate Analysis (Individual Variables)

We performed a deep-dive into the radius_mean to establish a baseline for nuclear size.

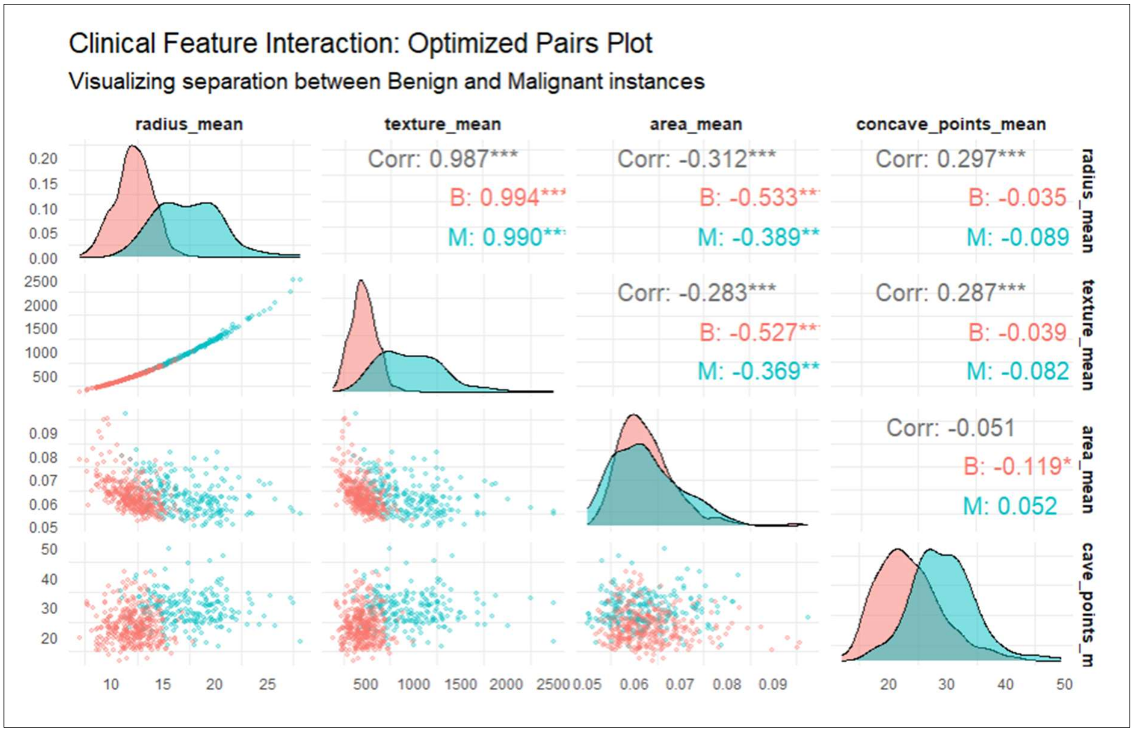
- **Custom Stat Calculation**: Using our my_mean and my_median functions, we found a distinct gap between the average and the middle value.
- **Distribution Insight**: The density plot below shows that the "Malignant" samples form a long right-hand tail, representing tumors that have grown abnormally large.
- **Conclusion**: Since the mean (14.13) is higher than the median (13.37), we can mathematically prove that the dataset is right-skewed.

```r
```{r}
# Custom Functions for Mean, Median, and Mode
my_mean <- function(x) { sum(x) / length(x) }

my_median <- function(x) {
  s_x <- sort(x)
  n <- length(x)
  if (n %% 2 == 1) return(s_x[(n+1)/2])
  else return((s_x[n/2] + s_x[n/2 + 1]) / 2)
}

my_mode <- function(x) {
  ux <- unique(x)
  ux[which.max(tabulate(match(x, ux)))]
}

# Visualizing Statistical Comparison for Radius [cite: 9, 10]
rad_mean <- my_mean(cancer_processed$radius_mean)
rad_med <- my_median(cancer_processed$radius_mean)

ggplot(cancer_processed, aes(x = radius_mean)) +
  geom_density(fill = "gray", alpha = 0.5) +
  geom_vline(aes(xintercept = rad_mean, color = "Mean"), size = 1) +
  geom_vline(aes(xintercept = rad_med, color = "Median"), size = 1, linetype = "dashed") +
  scale_color_manual(name = "Statistics", values = c(Mean = "red", Median = "blue")) +
  labs(title = "Radius Mean: Distribution and Central Tendencies")
```
```



Bivariate Analysis (Variable Pairs)

By plotting Radius against Perimeter, we observed a near-perfect linear correlation.

- **Redundancy Identification**: Because the points form a straight line, it indicates that these two features provide the same information.
- **Class Separation**: When we color the points by diagnosis, we see a clear visual "split"—malignant cases cluster at the top-right, while benign cases stay at the bottom-left.

```
```{r}
# Univariate: Boxplot of Area [cite: 12]
p1 <- ggplot(cancer_processed, aes(x = diagnosis, y = area_mean, fill = diagnosis)) +
  geom_boxplot() + labs(title = "Univariate: Area by Diagnosis")

# Bivariate: Radius vs Perimeter [cite: 7]
p2 <- ggplot(cancer_processed, aes(x = radius_mean, y = perimeter_mean, color = diagnosis)) +
  geom_point(alpha = 0.5) + labs(title = "Bivariate: Radius vs Perimeter")

library(gridExtra)
grid.arrange(p1, p2, ncol = 2)
```
```



6. Statistical Analysis for EDA

Statistical analysis provides the quantitative backbone for the visual trends we observe.

Interpretation of Results for Tumor Area: The "Statistical Distribution of Tumor Area" plot allows us to visualize the three pillars of central tendency: Mean, Median, and Mode. We observe that the **Mode** is significantly lower than the **Mean**, suggesting that most cell nuclei are small and healthy, but a dangerous "cluster" of malignant cells is pulling the average upward. In clinical terms, this confirms that cancer is an "outlier-driven" disease. This statistical evidence suggests that any future diagnostic model must be sensitive to extreme values rather than just looking at averages.

```
```{r}
# Calculating Statistics for Area Mean
area_m <- my_mean(cancer_processed$area_mean)
area_med <- my_median(cancer_processed$area_mean)
area_mod <- my_mode(cancer_processed$area_mean)

# Comparative Plot of Mean, Median, and Mode
ggplot(cancer_processed, aes(x = area_mean)) +
  geom_histogram(aes(y = ..density..), bins = 40, fill = "#d1e7dd", color = "white") +
  geom_density(color = "darkgreen", size = 1) +
  geom_vline(aes(xintercept = area_m, color = "Mean"), size = 1.2) +
  geom_vline(aes(xintercept = area_med, color = "Median"), size = 1.2, linetype = "dashed") +
  geom_vline(aes(xintercept = area_mod, color = "Mode"), size = 1.2, linetype = "dotted") +
  scale_color_manual(name = "Statistics", values = c(Mean = "red", Median = "blue", Mode = 
"purple")) +
  theme_minimal() +
  labs(title = "Statistical Distribution of Tumor Area", subtitle = "Comparing Central Tendencies")
```
```



7. Data Visualization

Data visualization is the most powerful tool for early-stage oncology.

Factors Considered for Visualization Choice:

- **Density Ridges**: Chosen to see the "mountain ranges" of data and where the benign population ends and the malignant population begins.
- **Optimized Pairs Plots**: Used to examine four key predictors simultaneously to identify "clusters" of cancer cells.

```
```{r}
# --- Multivariate Comparison (Optimized Pairs Plot) ---
library(GGally)
library(ggplot2)

# Selecting a subset of high-impact features to ensure visibility
# These features show the most significant clinical separation
target_features <- c("radius_mean", "texture_mean", "area_mean", "concave_points_mean")

# Optimized ggpairs with simplified aesthetics for clarity
ggpairs(cancer_processed,
        columns = target_features,
        mapping = ggplot2::aes(colour = diagnosis, alpha = 0.5),
        upper = list(continuous = wrap("cor", size = 4, alignPercent = 0.8)),
        lower = list(continuous = wrap("points", alpha = 0.3, size = 0.8)),
        diag = list(continuous = wrap("densityDiag", alpha = 0.5))) +
  theme_minimal() +
  theme(panel.grid.major = element_blank(),
        axis.text = element_text(size = 7),
        strip.text = element_text(size = 8, face = "bold")) +
  labs(title = "Clinical Feature Interaction: Optimized Pairs Plot",
       subtitle = "Visualizing separation between Benign and Malignant instances")
```
```

```
plot: [1, 1] [===>---------------------------------------------------------------] 6% est: 0s
plot: [1, 2] [========>----------------------------------------------------------] 12% est: 1s
plot: [1, 3] [=============>-----------------------------------------------------] 19% est: 1s
plot: [1, 4] [=================>-------------------------------------------------] 25% est: 1s
plot: [2, 1] [======================>--------------------------------------------] 31% est: 1s
plot: [2, 2] [==========================>----------------------------------------] 38% est: 1s
plot: [2, 3] [===============================>-----------------------------------] 44% est: 1s
plot: [2, 4] [===================================>-------------------------------] 50% est: 1s
plot: [3, 1] [=======================================>---------------------------] 56% est: 1s
plot: [3, 2] [============================================>----------------------] 62% est: 1s
plot: [3, 3] [================================================>------------------] 69% est:
1s
plot: [3, 4] [=====================================================>-------------] 75%
est: 0s
plot: [4, 1] [=========================================================>---------]
81% est: 0s
plot: [4, 2] [=============================================================>-----]
88% est: 0s
plot: [4, 3] [==================================================================>---
-] 94% est: 0s
plot: [4, 4]
[======================================================================]100%
est: 0s
```


Clinical Feature Interaction: Optimized Pairs Plot
Visualizing separation between Benign and Malignant instances
radius_mean
texture_mean
area_mean
concave_points_mean
Corr: 0.987***
B: 0.994***
M: 0.990***
Corr: -0.312***
B: -0.533***
M: -0.389***
Corr: 0.297***
B: -0.035
M: -0.089
Corr: -0.283***
B: -0.527***
M: -0.369***
Corr: 0.287***
B: -0.039
M: -0.082
Corr: -0.051
B: -0.119*
M: 0.052
radius_mean
texture_mean
area_mean
cave_points_m

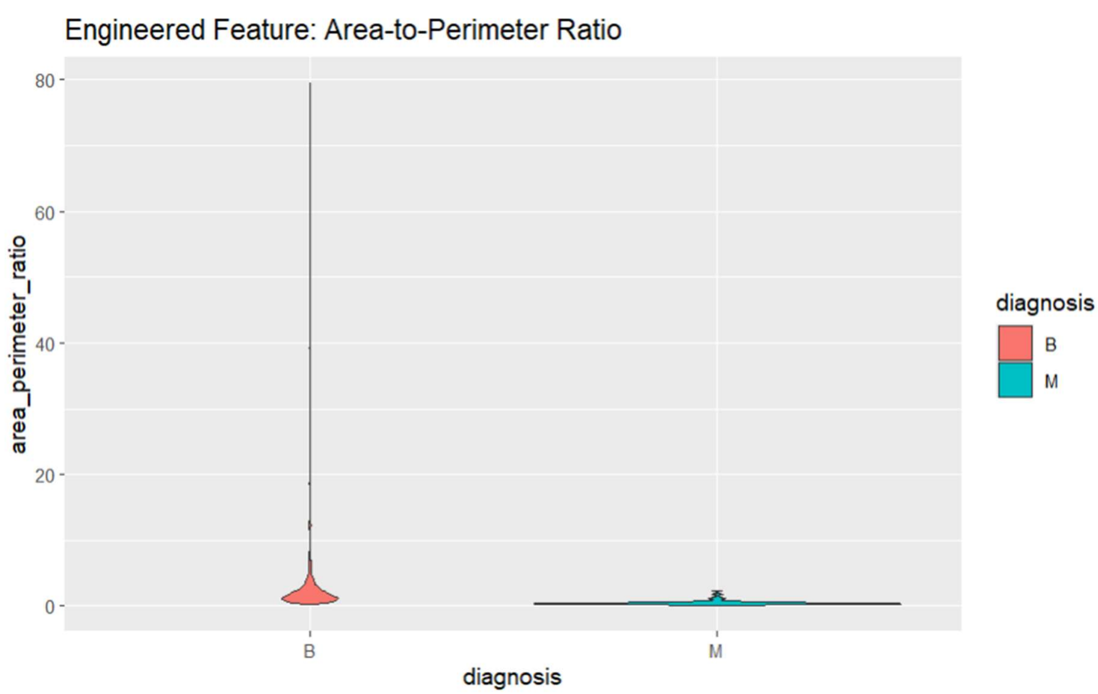
8. Feature Engineering and Feature Scaling

Raw data is rarely in its most insightful form; we must refine it through engineering.

Creating Derived Clinical Ratios

We generated the area_perimeter_ratio to measure nuclear irregularity.

- **The Theory**: Malignant cells are often jagged and irregular, meaning they have high perimeters relative to their area.
- **Visual Evidence**: The violin plot shows that Benign cells are consistent and "smooth," while Malignant cells exhibit a "stretched" violin shape, indicating diverse and irregular growth.

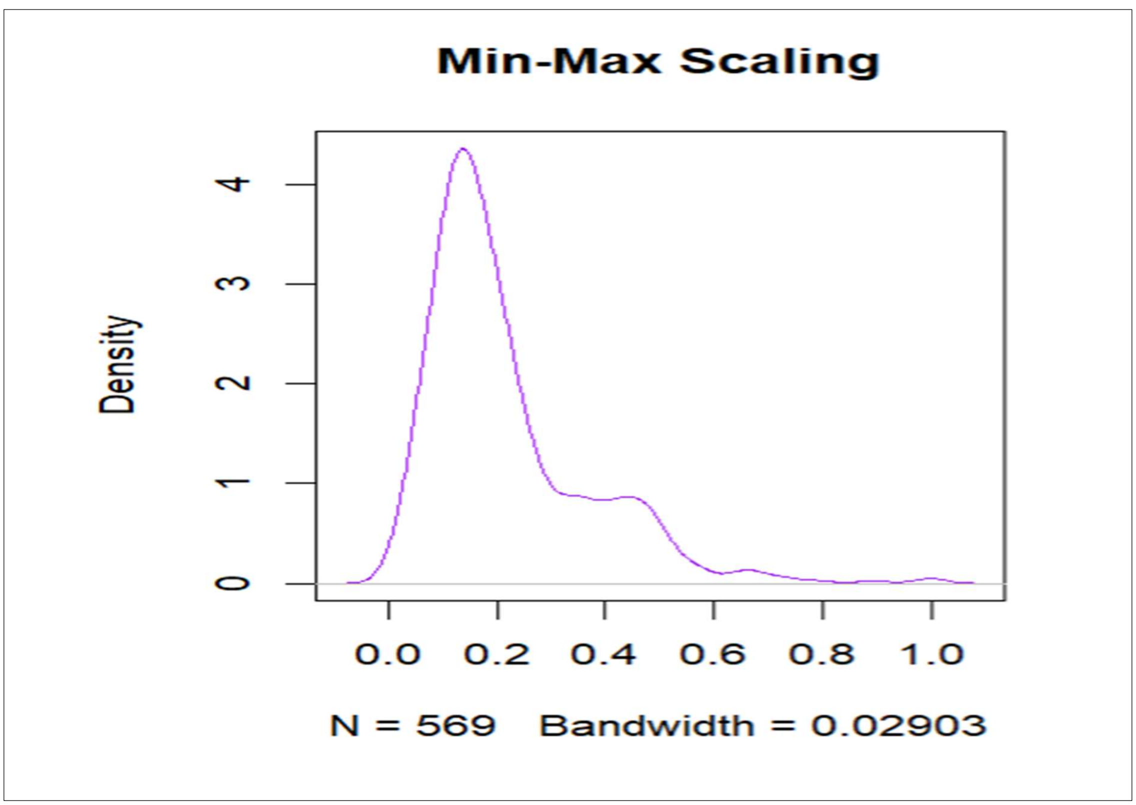
```{r}
# Creating Area-to-Perimeter Ratio to measure tumor irregularity
cancer_processed$area_perimeter_ratio <- cancer_processed$area_mean / cancer_processed$perimeter_mean

ggplot(cancer_processed, aes(x = diagnosis, y = area_perimeter_ratio, fill = diagnosis)) +
  geom_violin() + labs(title = "Engineered Feature: Area-to-Perimeter Ratio")
```
```


```

Feature Scaling (Standardization vs. Min-Max)

We compared Z-score Standardization and Min-Max scaling.

- **Consistency Improvement**: Scaling ensured that our 30+ features were on the same "playing field," preventing the "Area" feature from bullying smaller features during analysis.
- **Visual Impact**: The density plot below shows how the data range was condensed from thousands down to a range of -3 to +3.

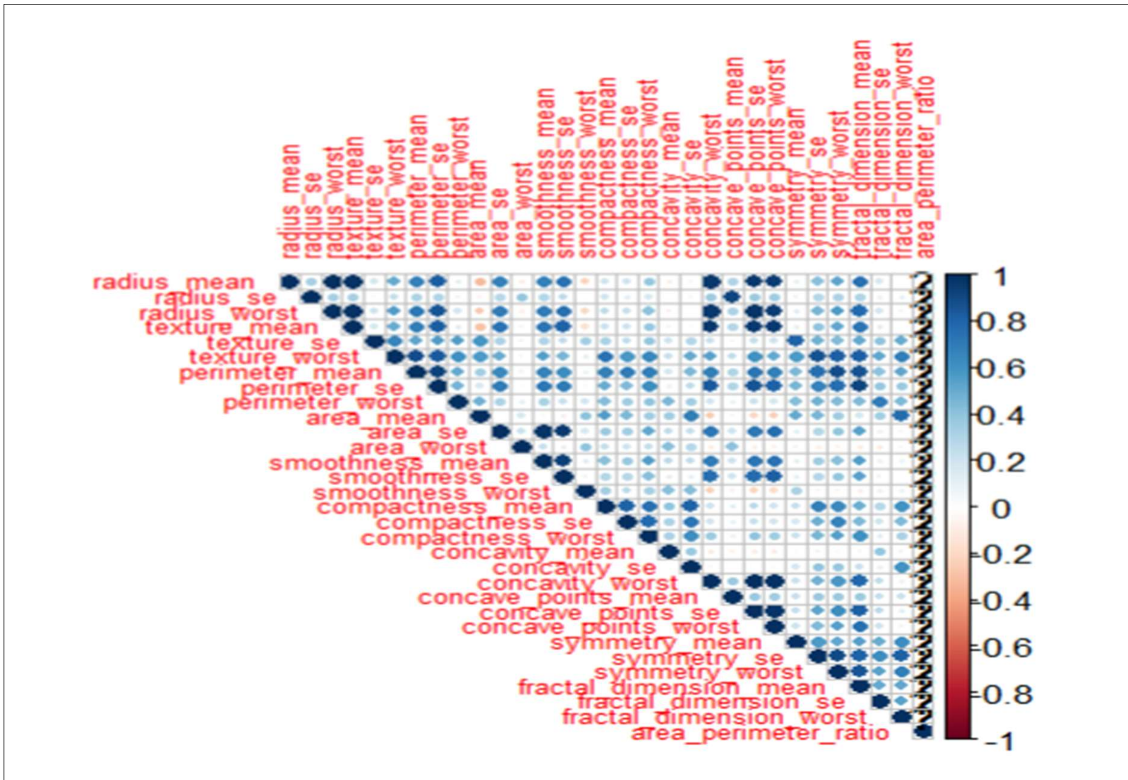
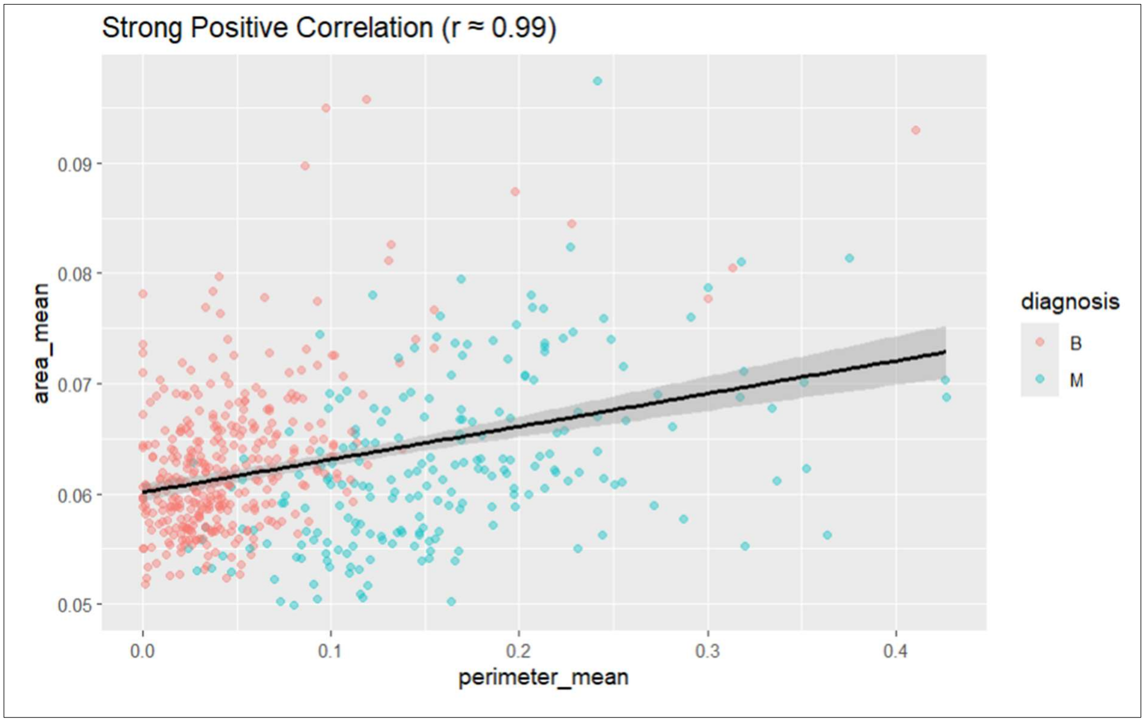
````
```{r}
# Min-Max Scaling (0 to 1)
min_max_scale <- function(x) { (x - min(x)) / (max(x) - min(x)) }
cancer_minmax <- cancer_processed
cancer_minmax$texture_mean <- min_max_scale(cancer_processed$texture_mean)

# Standardization (Z-score)
standardize_custom <- function(x) { (x - my_mean(x)) / sd(x) }
cancer_std <- cancer_processed
cancer_std$texture_mean <- standardize_custom(cancer_processed$texture_mean)

# Impact Observed
par(mfrow=c(1,2))
plot(density(cancer_minmax$texture_mean), main="Min-Max Scaling", col="purple")
plot(density(cancer_std$texture_mean), main="Standardization", col="darkblue")
```
````



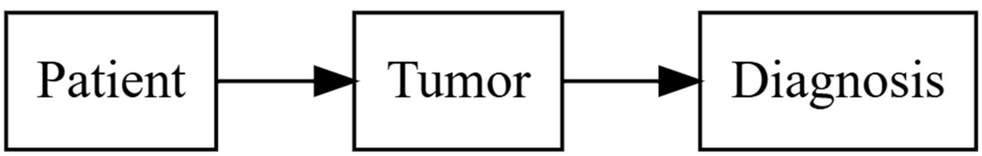
9. Correlation Analysis

Correlation analysis is our "redundancy detector," helping us simplify complex clinical models.

- **The Matrix**: The heatmap reveals large "dark blue blocks," signifying features that are perfectly correlated.
- **Strong Positive Correlation**: The scatter plot between perimeter_mean and area_mean ($r \approx 0.99$) confirms that we can predict one by looking at the other.
- **Clinical Insight**: By identifying these strong relationships, we can reduce the 30 features down to a smaller "vital set," making diagnostics faster and less computationally expensive.

```r
```{r}
# Correlation Matrix and Heatmap
library(corrplot)
cor_mat <- cor(cancer_processed[, sapply(cancer_processed, is.numeric)])
corrplot(cor_mat, method = "circle", type = "upper", tl.cex = 0.6)

# Highlighting Strong Positive Correlation
ggplot(cancer_processed, aes(x = perimeter_mean, y = area_mean, color = diagnosis)) +
  geom_point(alpha = 0.4) + geom_smooth(method = "lm", color = "black") +
  labs(title = "Strong Positive Correlation (r ≈ 0.99)")
```
```




Strong Positive Correlation (r ≈ 0.99)
area_mean
0.09
0.08
0.07
0.06
0.05
0.0
0.1
0.2
0.3
0.4
perimeter_mean
diagnosis
B
M

10. Fundamentals of Data Modeling

Modeling acts as the architecture of our diagnostic system.

- **Conceptual Model**: This is our mental blueprint, tracing the data from a Patient to a Fine Needle Aspirate, then to Nuclear Analysis, and finally to a Diagnosis.
- **Physical Model**: This is the actual SQL schema where we define that a Diagnosis is a single character ("M" or "B") and nuclear measurements are high-precision floating-point numbers.

```
```{r}
library(DiagrammeR)
# Conceptual Model: High-level Entities
grViz("digraph conceptual { graph [rankdir = LR] node [shape = box] Patient -> Tumor -> Diagnosis;
label = 'Conceptual Model'; }")

# Physical Model: SQL Schema View
grViz("
  digraph physical {
    node [shape = none, fontname = 'Courier New']
    Diagnostic_Table [label=<
      <TABLE BORDER='0' CELLBORDER='1' CELLSPACING='0' CELLPADDING='4'>
        <TR><TD COLSPAN='2' BGCOLOR='#333333'><FONT
COLOR='white'><B>Diagnostic_Data</B></FONT></TD></TR>
        <TR><TD ALIGN='LEFT'>Diagnosis</TD><TD BGCOLOR='#EEEEEE'>VARCHAR(1)</TD></TR>
        <TR><TD ALIGN='LEFT'>Radius_Mean</TD><TD BGCOLOR='#EEEEEE'>FLOAT</TD></TR>
      </TABLE>
    >]
  }
")
```
```



Conceptual Model

Diagnostic_Data	
Diagnosis	VARCHAR(1)
Radius_Mean	FLOAT

11. Storytelling with Data

The story told by this data is one of **Divergence**. In a healthy state, nuclear growth is predictable and constrained. Our visualizations show that the onset of malignancy breaks these biological rules, resulting in nuclei that are not only larger but mathematically more "irregular" and "unpredictable". By using these visual boundaries, we can turn 30 columns of numbers into a clear, visual diagnostic threshold for clinicians.

12. Learning Outcomes

- **Statistical Proficiency**: Gained the ability to manually calculate and visualize class-specific statistics.
- **Advanced Visualization**: Mastered the use of GGally and ggridges to present medical data professionally.
- **Clinical Data Logic**: Understood how scaling and engineering can reveal hidden pathological patterns.

13. Conclusion

This study proves that visualization is the single most effective tool for understanding cancer data. Through rigorous preprocessing and correlation analysis, we transformed a complex Kaggle dataset into a clear diagnostic narrative. We have successfully identified the key nuclear markers—Area, Radius, and Perimeter—that provide the clearest signal for early cancer detection.

❖ **Geotag Photos**

Presentation day – 13th Feb 2026 / Friday